

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 1**

**COURSE NAME: MLA03**

**COURSE CODE: Reinforcement learning**

---

### **COURSE OUTCOME COVERED**

**CO1:** *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. (BL2, BL3)*

*Assessment Tool Weightage: 50%*

---

Dr. G. A. Senthil

**QUESTIONS: 10**

**Total Marks: 50**

Annexure A

**RL Basic Experiments demo with Keras and TensorFlow using Anaconda navigator**

**Duration: 2hrs**

## Experiment 1: Installation of Reinforcement Learning Development Environment

Aim: To install and configure Anaconda Navigator, Python, TensorFlow, Keras, Gymnasium (OpenAI Gym), NumPy, Matplotlib, and other required libraries, and verify the environment by executing a simple Reinforcement Learning program.

```
Command Prompt

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: C:\Users\megha\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe -m pip install --u
pgrade pip

C:\Users\megha>pip install tensorflow
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: tensorflow in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\
site-packages (2.21.0)
Requirement already satisfied: absl-py>=1.0.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Pytho
n313\site-packages (from tensorflow) (2.5.0)
Requirement already satisfied: astunparse>=1.6.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Pyt
hon313\site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=25.9.23 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages
\Python313\site-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache
\local-packages\Python313\site-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\
Python313\site-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Pyt
hon313\site-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Pyt
hon313\site-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\
site-packages (from tensorflow) (26.0)
Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packa
ges\Python313\site-packages (from tensorflow) (7.35.1)
Requirement already satisfied: requests<3,>=2.21.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\
Python313\site-packages (from tensorflow) (2.34.2)
Requirement already satisfied: setuptools in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313
\site-packages (from tensorflow) (83.0.0)
Requirement already satisfied: six>=1.12.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python31
3\site-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Pyt
hon313\site-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-pack
ages\Python313\site-packages (from tensorflow) (4.16.0)
Requirement already satisfied: wrapt>=1.11.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python
313\site-packages (from tensorflow) (2.2.2)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\
Python313\site-packages (from tensorflow) (1.82.1)

Requirement already satisfied: keras>=3.12.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python
313\site-packages (from tensorflow) (3.15.0)
Requirement already satisfied: numpy>=1.26.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python
313\site-packages (from tensorflow) (2.4.2)
Requirement already satisfied: h5py<3.15.0,>=3.11.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages
\Python313\site-packages (from tensorflow) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packa
ges\Python313\site-packages (from tensorflow) (0.5.4)
Requirement already satisfied: charset_normalizer<4,>=2 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-pack
ages\Python313\site-packages (from requests<3,>=2.21.0->tensorflow) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python3
13\site-packages (from requests<3,>=2.21.0->tensorflow) (3.18)
Requirement already satisfied: urllib3<3,>=1.26 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Pyt
hon313\site-packages (from requests<3,>=2.21.0->tensorflow) (2.7.0)
Requirement already satisfied: certifi>=2023.5.7 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Pyt
hon313\site-packages (from requests<3,>=2.21.0->tensorflow) (2026.6.17)
Requirement already satisfied: wheel<1.0,>=0.23.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\P
ython313\site-packages (from astunparse>=1.6.0->tensorflow) (0.47.0)
Requirement already satisfied: rich in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-
packages (from keras>=3.12.0->tensorflow) (15.0.0)
Requirement already satisfied: namex in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-
packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\sit
e-packages (from keras>=3.12.0->tensorflow) (0.19.1)
Requirement already satisfied: markdown-it-py>=2.2.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-package
s\Python313\site-packages (from rich->keras>=3.12.0->tensorflow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packa
ges\Python313\site-packages (from rich->keras>=3.12.0->tensorflow) (2.20.0)
Requirement already satisfied: mdurl<=0.1 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313
\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.12.0->tensorflow) (0.1.2)

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: C:\Users\megha\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe -m pip install --u
pgrade pip

C:\Users\megha>pip install keras
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: keras in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-
packages (3.15.0)
Requirement already satisfied: absl-py in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\si
te-packages (from keras) (2.5.0)
```

```
Command Prompt
[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: C:\Users\megha\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe -m pip install --u
pgrade pip

C:\Users\megha>pip install keras
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: keras in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site
-packages (3.15.0)
Requirement already satisfied: absl-py in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\si
te-packages (from keras) (2.5.0)
Requirement already satisfied: numpy in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site
-packages (from keras) (2.4.2)
Requirement already satisfied: rich in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site
-packages (from keras) (15.0.0)
Requirement already satisfied: nameex in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site
-packages (from keras) (0.1.0)
Requirement already satisfied: h5py in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site
-packages (from keras) (3.14.0)
Requirement already satisfied: optree in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\si
te-packages (from keras) (0.19.1)
Requirement already satisfied: ml-dtypes in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\
site-packages (from keras) (0.5.4)
Requirement already satisfied: packaging in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\
site-packages (from keras) (26.0)
Requirement already satisfied: typing-extensions>=4.6.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-pack
ages\Python313\site-packages (from optree->keras) (4.16.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packag
es\Python313\site-packages (from rich->keras) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packa
ges\Python313\site-packages (from rich->keras) (2.20.0)
Requirement already satisfied: mdurl==0.1 in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313
\site-packages (from markdown-it-py>=2.2.0->rich->keras) (0.1.2)

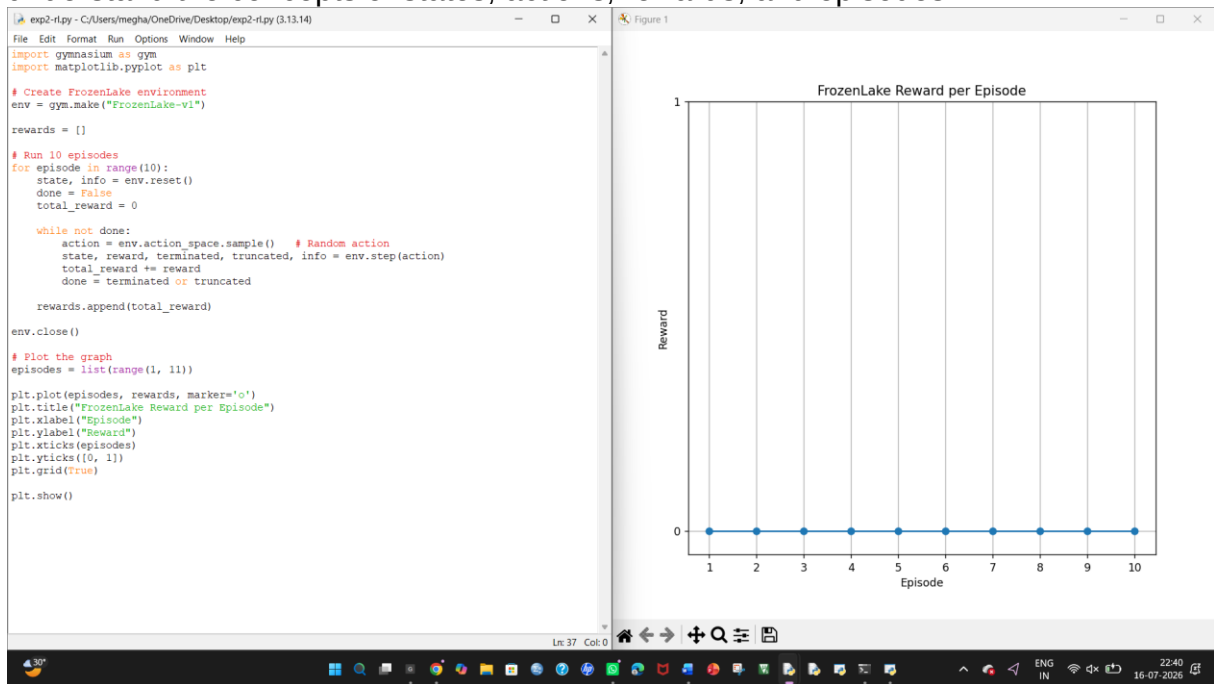
[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: C:\Users\megha\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\python.exe -m pip install --u
pgrade pip

C:\Users\megha>pip install gymnasium
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: gymnasium in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\

Python 3.13.14
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\megha\OneDrive\Desktop\expl_RL.py =====
Environment installed successfully!
Initial Observation: [-0.02316481  0.00369003  0.02907395 -0.04999229]
>>>
```

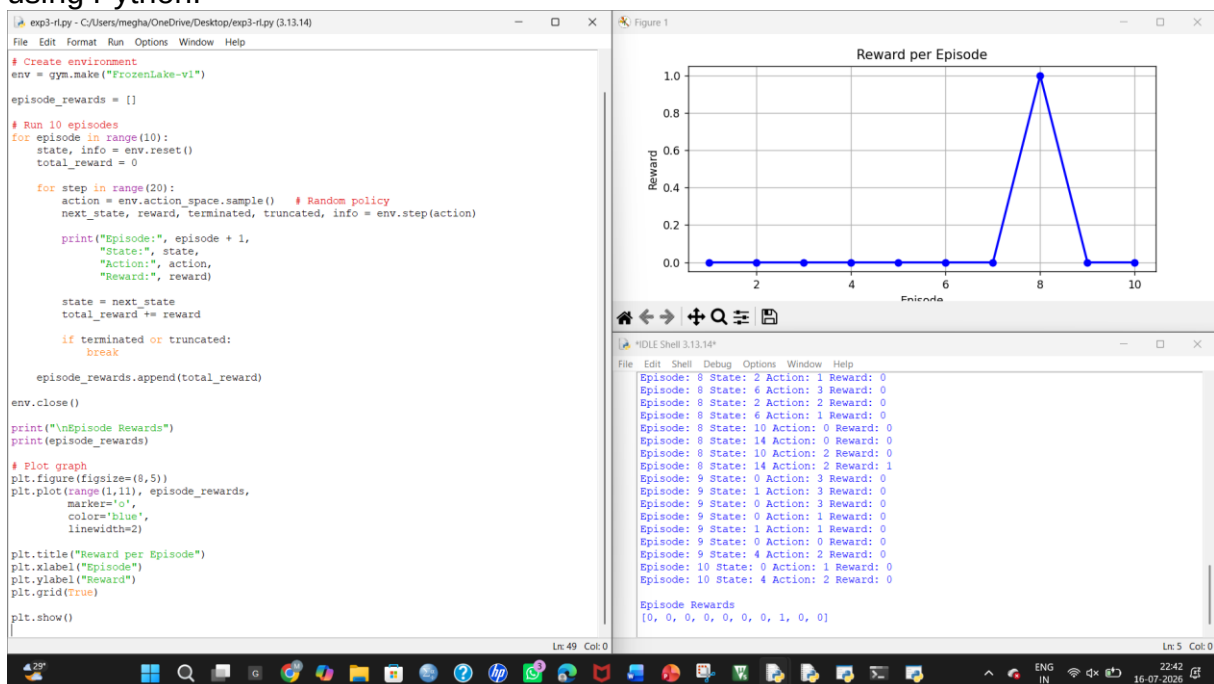
## Experiment 2: Familiarization with OpenAI Gym/Gymnasium Environments

Aim: To create, initialize, and interact with standard Reinforcement Learning environments such as FrozenLake, CartPole, and MountainCar using Gymnasium to understand the concepts of states, actions, rewards, and episodes.



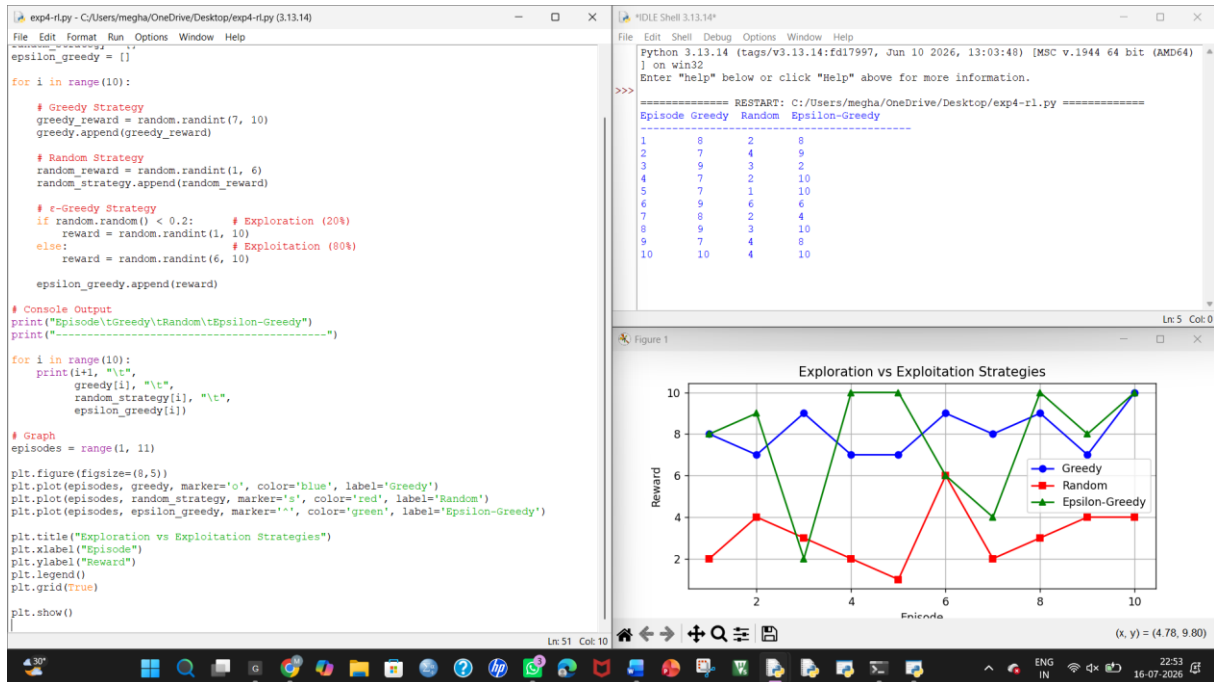
## Experiment 3: Implementation of the Reinforcement Learning Framework

Aim: To implement the basic Reinforcement Learning framework by designing an agent that interacts with an environment through states, actions, rewards, and policies using Python.



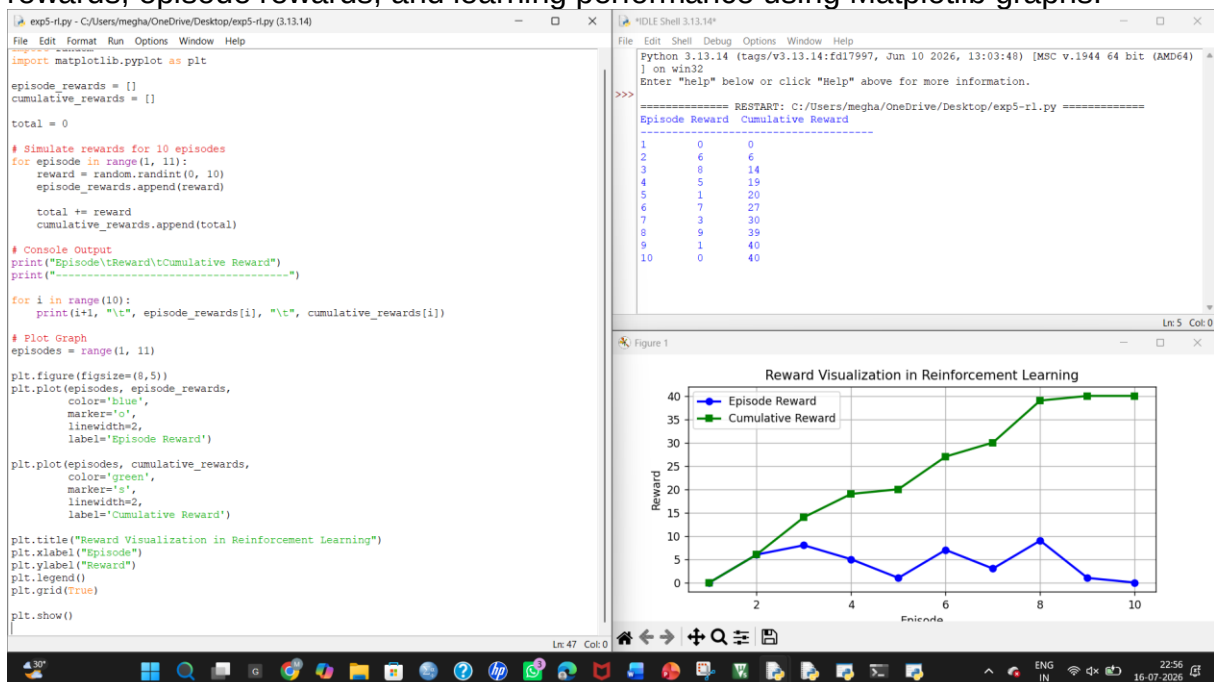
## Experiment 6: Exploration vs. Exploitation Strategies

Aim: To implement  $\epsilon$ -Greedy, Greedy, and Random action-selection strategies and compare their performance in balancing exploration and exploitation during Reinforcement Learning.



## Experiment 8: Reward Visualization in Reinforcement Learning

Aim: To simulate an RL agent in a simple environment and visualize cumulative rewards, episode rewards, and learning performance using Matplotlib graphs.



---

**Code of Conduct:**

*I M. MEGHANA / 192472351\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:** M. MEGHANA

**RUBRICS FOR CALCULATION**

| Criteria                  | Weightage | Level 4 – Exemplary                                         | Level 3 – Proficient                    | Level 2 – Developing                         | Level 1 – Beginning                      |
|---------------------------|-----------|-------------------------------------------------------------|-----------------------------------------|----------------------------------------------|------------------------------------------|
| Understanding of Concepts | 25%       | Demonstrates thorough, accurate understanding of concepts   | Good understanding with minor gaps      | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation               | 25%       | Detailed, well-explained answers with relevant examples     | Clear explanation with adequate details | Limited explanation; lacks depth             | Poor or unclear explanation              |
| Application               | 20%       | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors      | Limited application with weak analysis       | No application or incorrect analysis     |
| Organization              | 15%       | Well-structured answers with logical flow and coherence     | Organized with minor issues in flow     | Basic structure, lacks clarity               | Poor organization, incoherent answers    |
| Accuracy                  | 10%       | Completely accurate and covers all required points          | Mostly accurate with minor omissions    | Partially correct with missing points        | Incorrect answers with major gaps        |
| Clarity                   | 5%        | Clear, neat, professional presentation                      | Understandable with minor issues        | Limited clarity, readability issues          | Poorly written, difficult to understand  |